Sai Sena Chinnakonduru

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Education

Indiana University Bloomington **Aug 2022 - May 2024**
Master of Science, Data Science - Algorithms, Artificial Intelligence, Software Engineering, Machine Learning, Deep Learning ***GPA: 4.0/4.0***

National Institute of Technology, Kurukshetra **Aug 2015 – May 2019**
Bachelor of Technology, Civil Engineering - Mathematics, Linear Programming, Statistics,Data Structures ***GPA: 3.6/4.0***

Skills

Programming Languages: Python,SQL,R,PySpark
Machine Learning: Regression Algorithms, NLP(Natural Language Processing), Decision Tree, Bayesian Regression
Deep Learning and AI: Computer Vision, Transformers, Generative AI (Gen AI), Large Language Models(LLMs), Autoencoders
Data Science: Statistics and Probability, Exploratory Data Analysis, Excel, Tableau, Power BI
Frameworks: Tensorflow, PyTorch, Huggingface Transformers, Numpy, Scikit-learn, NLTK(Natural Language Toolkit), Flask
Databases: PostgreSQL, MySQL, MongoDB
Tools & Technologies: Hadoop, Spark, MapReduce, Postman, Git,Azure,Jira

Experience

Senior Data Scientist| **Tredence Inc.** **Jan 2021 - Jul 2022**
Deep Learning, Python, Optimization, Linear Programming, Time-Series Forecasting, Azure,PySpark,Flask *Bangalore, India*

- Constructed a CNN model to classify the severity of Rheumatoid Arthritis based on thermal images, achieving an accuracy of **95%**.
- Led network optimization model design, reducing penalty costs by **$2M** annually through improved shipment recommendations.
- Engineered a fbprophet-based forecasting model for optimizing inventory, achieved a MAPE of **14%** with external indicators.
- Built Azure Data Factory pipelines to streamline data extraction and transformation, supporting model execution and dashboards.
- Created Flask-based APIs for automated promotional campaign reporting, reducing manual processing time from **2 days to 1 minute**.

Junior Machine Learning Engineer | **Tredence Inc.** **June 2019 - Dec 2020**
Python, SQL, Machine Learning, Decision Tree, EDA, Flask *Bangalore, India*

- Optimized SKU-warehouse mapping network, reducing lead times from warehouse to stores from **13 days to 8 days**.
- Developed a Flask based financial tool, integrating APIs to streamline profitability reporting, eliminating **1.5 days** of manual effort.
- Constructed a decision tree model with **75%** accuracy to analyze customer return rates, aiding targeted marketing strategies.
- Analyzed store operational metrics to identify factors impacting customer experience and provided actionable insights.
- Utilized Tableau for publishing customer satisfaction metrics through data visualization dashboards, streamlining decision-making.

Projects

Weighted Grouped Query Attention - Transformers
PyTorch, Transformers, KV caching, Efficient Attention, Distributed Training | ***Code***

- Investigated weighted grouped query attention, alternate approach to grouped multi-query attention in transformers within the T5 language model, improving rouge-1 score by **0.1 to 0.3** on CNN dailymail dataset.
- Finetuned the model using PyTorch's Fully Sharded Data Parallel (FSDP) reducing the training time by **4x**.

Using LLMs for node classification in graphs
PyTorch, GNN, LLM

- Analyzed OGBN-arxiv dataset from the Microsoft Academic Graph with **100k** nodes and **1.5 million** edges.
- Employed different LLMs-Voyage,OpenAI, T5, and Gemini to generate embeddings of text-attributed nodes in graphs for training GNN.
- Conducted experiments with various GNN configurations for node classification, and achieved an accuracy of **72%**.

Grade 4 and 5 Math Worksheet Image Classification
Python, Tensorflow, Computer Vision

- Developed a computer vision pipeline to analyze math worksheets, used EfficientNet for feature extraction and preprocessing images.
- Utilized a hierarchical SVM classifier on extracted features to categorize worksheets with drawings or text, achieving **91%** accuracy.

Speech Denoising and Classification, Model Compression
Python, Tensorflow, Deep Learning, Neural Networks, Audio Processing, Model Compression | ***Code***

- Trained DL models (CNN, GRU, LSTM) to denoise speech recordings, achieving SNR of **12 dB**.
- Designed a Siamese network for speaker verification, achieving **74%** accuracy in identifying if speeches were by the same speaker.
- Compressed the model to classify MNIST digits using SVD from **19MB to 841KB** with **1%** loss in accuracy.

Image Generation using RNN and GAN
Python, Tensorflow, Deep Learning, RNN, LSTM | ***Code***

- Built LSTM model for image generation tasks, specifically to reconstruct the bottom half of images from the top half.
- Trained a conditional GAN model to synthesize high-quality MNIST digits from random noise.

Accomplishments

- Supervised a class of **160** students as an Associate Instructor for the Elements of Artificial Intelligence graduate course.
- Conducted office hours, and graded assignments as a Teaching Assistant for Machine Learning graduate class of **35** students.